

CAPE SUCKLING BUOY 46082, AK, USA

WMO: 997204

Lat: 59.681N

Lon: 143.372W

Elev: 0

StdP: 14.70

Time Zone: -10.00 (W10)

Period: 05-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	24.7	26.8	N/A	N/A	N/A	N/A	N/A	N/A	47.9	35.4	42.9	37.4	16.2	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.7	59.8	N/A	59.1	N/A	58.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	9.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 42.0	38.4	34.9	DB	22.8	61.2	2.7	2.0	20.9	62.7	19.3	63.8	17.8	65.0	15.9	66.4	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	37.1	37.2	36.6	40.3	45.4	50.9	54.9	55.6	52.4	46.5	40.7	39.3	(6)	
	DBStd	7.72	5.07	4.40	4.17	2.78	3.26	3.11	2.66	1.73	2.45	3.06	4.12	3.61	(7)	
	HDD50	2363	398	358	416	292	148	24	1	0	6	110	278	332	(8)	
	HDD65	7374	863	778	881	742	609	422	313	291	378	572	728	797	(9)	
	CDD50	464	0	0	0	0	5	52	153	174	78	2	0	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	15.1	18.2	16.4	17.0	14.5	12.3	11.3	10.4	12.0	15.7	16.2	18.4	19.0	(14)	
	PrecAvg	63.9	6.2	5.2	4.1	3.4	3.3	3.0	4.3	5.9	7.7	7.5	6.5	6.8	(15)	
(16) Precipitation	PrecMax	77.5	11.4	7.9	8.3	6.1	5.9	5.7	6.1	9.2	12.6	11.5	11.9	10.5	(16)	
	PrecMin	43.4	1.9	2.1	1.8	1.7	1.1	1.1	2.0	2.0	3.9	2.8	2.8	3.1	(17)	
	PrecStd	7.3	2.2	1.7	1.7	1.1	1.4	0.9	1.0	2.1	2.2	2.4	2.4	2.0	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.3	46.2	44.6	49.9	55.3	59.6	61.4	60.7	58.4	53.2	49.4	47.1	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	45.3	44.5	43.6	47.0	53.0	58.2	60.1	59.7	57.5	51.9	48.3	45.7	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	44.1	43.8	42.8	45.0	51.3	56.7	59.3	59.0	56.9	51.1	47.2	44.8	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	43.0	42.7	42.0	43.7	50.1	55.4	58.6	58.3	56.0	50.4	46.1	43.9	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	4.6	4.2	4.3	3.4	2.8	2.6	2.6	2.7	2.9	3.6	4.8	4.2	(35)	
		MCDBR	3.3	4.3	3.7	4.2	4.0	3.4	3.2	3.5	3.0	3.3	4.5	3.8	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.289	0.317	0.331	0.376	0.381	0.381	0.376	0.378	0.354	0.335	0.299	N/A	(40)	
	taud		2.199	2.333	2.369	2.256	2.323	2.387	2.450	2.442	2.469	2.345	2.168	N/A	(41)	
	Ebn at Noon		156	220	255	260	268	270	269	259	246	207	144	N/A	(42)	
	Edh at Noon		14	21	27	36	36	34	32	30	24	20	14	N/A	(43)	
(44) All-Sky Solar Radiation	RadAvg		90	290	711	1250	1590	1636	1422	1093	749	360	128	55	(44)	
	RadStd		12	46	82	102	121	104	123	150	97	40	16	6	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page